CCSU BCA | SEMESTER I

Indian Knowledge System

Knowledge traditions, mathematics, science and society

COURSE CODE	BCA-1003T | Archive label: 19009
SEMESTER	BCA Semester I
SYLLABUS	CCSU BCA 2025-26 / NEP

This guide presents the syllabus as a set of knowledge traditions, methods and historical examples. Keep claims tied to their context and explain how a concept is used, rather than treating a historical analogy as identical to a modern technology.

Targeting 80+ is a study goal, not a guaranteed result. Use these notes with class material, the current syllabus and timed paper practice.

How to use this guide

Learn the short unit summary first. Then answer each question without looking, compare your answer with the model, and rewrite it in your own words. For calculations and programs, write every step and test with a small example.

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Unit I | Introduction to IKS

IKS surveys bodies of learning and ways of organizing knowledge across texts, philosophy, education, arts and practical skills.

Core ideas to remember

- The four Vedas are Rigveda, Yajurveda, Samaveda and Atharvaveda.
- The six Vedangas are Shiksha, Kalpa, Vyakarana, Nirukta, Chandas and Jyotisha.
- The syllabus names Vedic schools such as Samkhya, Yoga, Nyaya, Vaisheshika, Mimamsa and Vedanta, and non-Vedic traditions such as Charvaka, Buddhist and Jain thought.
- Itihasa commonly refers here to the Ramayana and Mahabharata; Puranic and niti literature have distinct genres and purposes.
- Caturdasa Vidyasthana and 64 Kalas are syllabus concepts for classifying knowledge and arts; explain them in the course's terms.

Important questions with model answers

Question: What are the Vedas and Vedangas?

Model answer

The four Vedas are Rigveda, Yajurveda, Samaveda and Atharvaveda. The Vedangas are six supporting disciplines: Shiksha concerns sound and pronunciation; Kalpa concerns ritual procedures; Vyakarana concerns grammar; Nirukta explains difficult words; Chandas studies metre; and Jyotisha concerns timekeeping and calendrical matters. Together they show how textual knowledge was studied through different specialized methods.

Question: Briefly distinguish Vedic and non-Vedic philosophical schools.

Model answer

The syllabus groups Samkhya, Yoga, Nyaya, Vaisheshika, Purva-Mimamsa and Vedanta as Vedic schools, while Charvaka, Buddhist and Jain traditions are presented as non-Vedic. Their views differ on knowledge, reality, self, causation and liberation. A strong exam answer names the schools and gives one central concern of each instead of suggesting that all Indian philosophy has one single position.

Unit II | Foundations for science and technology

This unit relates linguistic analysis, number representation, measurement and theories of knowledge to the history of science and technology.

Core ideas to remember

- Panini's Ashtadhyayi is a highly organized grammatical work; comparisons with computation are useful as an analogy about rules and formal structure, not as proof that it is a modern programming language.
- Place value and zero make positional number representation compact and support arithmetic.
- Bhuta-Sankhya and Katapayadi encode numbers through words or symbols using conventional mappings.
- Pingala's work on metre describes patterns of short and long syllables that can be represented combinatorially; connect this carefully to binary patterns.
- In Vaisheshika terminology, prameya is an object of knowledge, pramana a means of valid knowledge, and samshaya a state of doubt or uncertainty.

Important questions with model answers

Question: Explain the significance of place value and zero.

Model answer

In a positional decimal system, a digit's value depends on both the digit and its position: 352 means 3 hundreds, 5 tens and 2 ones. Zero can act as a number and as a placeholder, distinguishing values such as 205 and 25. Positional representation makes written arithmetic more systematic than a purely additive notation. In an exam answer, use a small place-value example.

Question: What are pramana, prameya and samshaya?

Model answer

Pramana is a means or source of valid knowledge; prameya is the object or subject to be known; samshaya is doubt about what is the case. In a simple example, a student sees a distant shape and is unsure whether it is a person or a post. Careful observation can reduce the doubt. Use the terms as concepts in the syllabus and explain their relationship in a knowledge process.

Unit III | Indian mathematics and astronomy

The unit introduces mathematical practices and astronomical traditions through texts, calculations, calendars and instruments.

Core ideas to remember

- The Sulba Sutras preserve geometric procedures associated with altar construction; discuss the mathematical method and its setting.
- Aryabhata's Aryabhatiya is a major work on mathematics and astronomy; Brahmagupta and Bhaskara are also important historical mathematicians.
- The syllabus includes arithmetic, geometry, algebra, trigonometry, pi and Pingala's Chandah Sastra.
- A Panchanga organizes calendrical information using tithi, vara, nakshatra, yoga and karana.
- Jantar Mantar observatories used large instruments for observing and calculating astronomical positions; name the historical context when discussing them.

Important questions with model answers

Question: What is a Panchanga? Name its five parts.

Model answer

Panchanga is a traditional Indian calendar framework. Its five limbs are tithi, a lunar day; vara, the weekday; nakshatra, a lunar mansion; yoga, a calculated angular relationship; and karana, half of a tithi. It combines several astronomical measures to support calendrical and ritual decisions. State each term and its role rather than calling it simply a list of dates.

Question: Describe the contribution of the Sulba Sutras to geometry.

Model answer

The Sulba Sutras contain geometric rules and procedures associated with constructing ritual spaces and altars. They show practical geometry through measurement, transformation of shapes and spatial construction. A careful answer situates the material in its historical and ritual setting, explains one example of a geometric procedure, and avoids assuming that a short traditional rule is equivalent to a modern proof unless the specific text supports that comparison.

Unit IV | Indian science and technology

This unit surveys technical knowledge in materials, metalworking, painting, urban planning, architecture and occupational arts.

Core ideas to remember

- Discuss metals such as copper, gold, zinc, mercury, lead and silver through their historical uses and production methods.
- Iron and steel history can include furnaces, alloying practices, tools and trade; distinguish documented techniques from broad claims.
- Dyes and painting technologies combine material knowledge with craft practice.
- Town planning and temple architecture involve layout, materials, structural forms and local traditions.
- Vastu Shastra is a body of architectural principles and texts; describe its historical concepts without presenting every claim as modern engineering evidence.

Important questions with model answers

Question: How can ancient Indian metalworking be discussed in an exam answer?

Model answer

Start by naming the material and its use, then describe the knowledge involved: ore selection, heating, smelting, shaping, joining or finishing. Mention iron and steel, copper, zinc or other metals covered in class. Connect the work to artisans, trade and social needs. Avoid unsupported statements about universal superiority; a careful answer gives the example, time or region when the source is known.

Question: Describe major elements of Indian temple architecture.

Model answer

A temple is both a sacred space and a designed structure. Common elements discussed in many traditions include a sanctum, a tower or superstructure, halls, gateways, sculpture and a planned relationship to the surrounding site. Regional forms vary, so identify the style or region when known. Explain how layout, symbolism, materials and craft work together rather than describing all temples as identical.

Unit V | Humanities and social sciences

The final unit brings together health, food, sleep, yoga, psychology, ethics, governance and public administration.

Core ideas to remember

- The syllabus asks how food, sleep, water and daily practice relate to health and wellbeing in Ayurveda and yoga traditions.
- Yoga is a set of practices and philosophies; name the specific path or practice rather than treating yoga as only posture.
- The Triguna framework uses sattva, rajas and tamas to discuss qualities of mind and nature in some philosophical traditions.
- Ramayana and Arthashastra offer different textual perspectives on governance, duty, administration and statecraft.
- Use historical context and distinguish a text's normative ideas from present-day constitutional or administrative rules.

Important questions with model answers

Question: Explain the Triguna framework in brief.

Model answer

The Triguna framework describes three qualities: sattva is associated with clarity and balance; rajas with activity and desire; and tamas with inertia or obscurity. In the syllabus it supports an Indian approach to psychology and wellbeing. A sound answer explains the terms as a conceptual framework within particular traditions, not as measured modern personality categories.

Question: How can the Arthashastra be used to discuss governance?

Model answer

The Arthashastra is a classical text associated with statecraft and administration. It can be discussed through its treatment of the ruler, officials, revenue, law, diplomacy and security. Explain the text's historical setting and its practical orientation. Avoid equating its prescriptions directly with contemporary democratic institutions; instead compare its questions about public order and administration with modern governance carefully.

Paper pattern and topic analysis

The current syllabus assigns 25 internal plus 75 external marks. The CCSUStudy BCA archive lists a December 2025 BCA Semester I IKS paper (label 19009). The full paper text and section structure were not accessible for independent checking, so this guide does not claim a verified question pattern.

Only the current 2025 paper listing is identified for BCA IKS. No earlier BCA IKS papers appear in the checked five-year Semester I archive. This prevents a meaningful year-to-year repetition analysis. The five modules have six lectures each in the syllabus, so use that balanced design as a revision guide, not as a marks-weight prediction.

What to prioritize

- Memorize the names and short definitions: four Vedas, six Vedangas, philosophical schools, five Panchanga limbs and key knowledge terms.
- Prepare one clear example per module: Panini, place value, Pingala, Sulba geometry, Jantar Mantar, metallurgy, yoga or Arthashastra.
- Answer historical questions with context, a specific example and a measured explanation of significance.
- Separate a traditional framework, a modern analogy and a scientifically demonstrated claim.

Last-five-year archive check

The archive lists one current BCA IKS paper for 2025 (label 19009). No BCA IKS paper for 2021-2024 was found in the checked CCSUStudy BCA Semester I archive. The syllabus and official IKS learning material therefore carry most of the weight in these notes.

How to read the evidence

Unit coverage follows the 2025-26 CCSU BCA syllabus reproduction. Paper-year availability comes from the CCSUStudy archive, which identifies itself as unofficial. Model answers are original summaries; no multi-year topic frequency is claimed.

Learning resources and source list

The paper archive is a discovery aid, not the university's official syllabus. The current course titles and units below follow the 2025-26 CCSU syllabus reproduced by an affiliated college. Older paper titles are identified as legacy or related material wherever they differ.

CCSU BCA syllabus and unit list (2025-26 reproduction)

CCSUStudy BCA question-paper archive

December 2025 IKS paper listing

Ministry of Education IKS Division textbook and course material

IKS Division curated learning resources

Coverage note: no question bank can predict the exact paper. Treat the priority list as a revision map and prepare every listed unit.